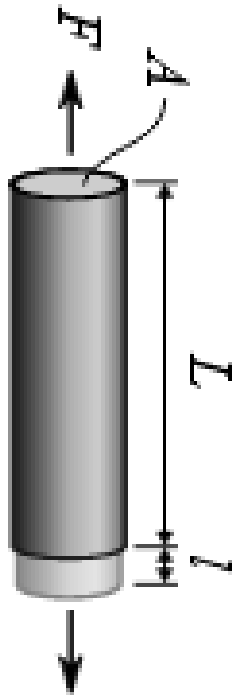


STM Exercise 2

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$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m²]

$$\varepsilon = \frac{\Delta L}{L_0}$$

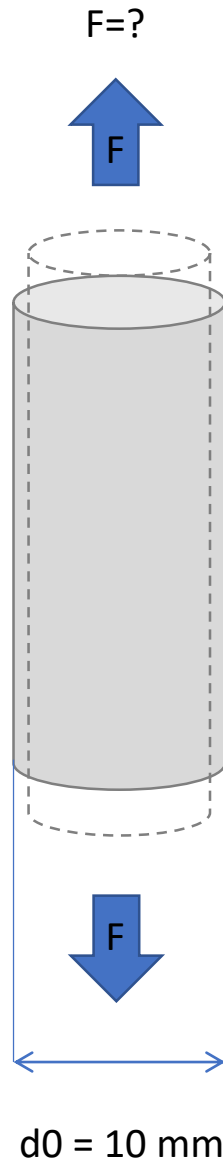
where

- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

$$E = \frac{\sigma}{\varepsilon}$$

Exercise 1: You have a cylindrical component made of an Al alloy (10 mm diameter, $Y_S = 150 \text{ MPa}$): which is the maximum load you can apply on this component without having plastic deformation? Consider a safety factor of $n = 2$.

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Yield strength $Y_S (\sigma_y) = 150 \text{ MPa}$
Safety factor = 2

No plastic deformation $\rightarrow \sigma \leq \sigma_y$
Safety factor = 2 $\rightarrow \sigma \leq \sigma_y / 2$

$$\sigma_{\max} = \sigma_y / 2 = 150 / 2 = 75 \text{ MPa}$$

$$A = \pi r^2 = 78.5 \text{ mm}^2$$

$$F_{\max} = \sigma_{\max} * A = 75 * 78.5 = 5890 \text{ N}$$

Poisson's ratio, ν

- Poisson's ratio, ν :**

$$\nu = \frac{-\epsilon_x}{\epsilon_z} = \frac{-\epsilon_y}{\epsilon_z}$$

(Isotropic materials only)

metals: $\nu \sim 0.33$

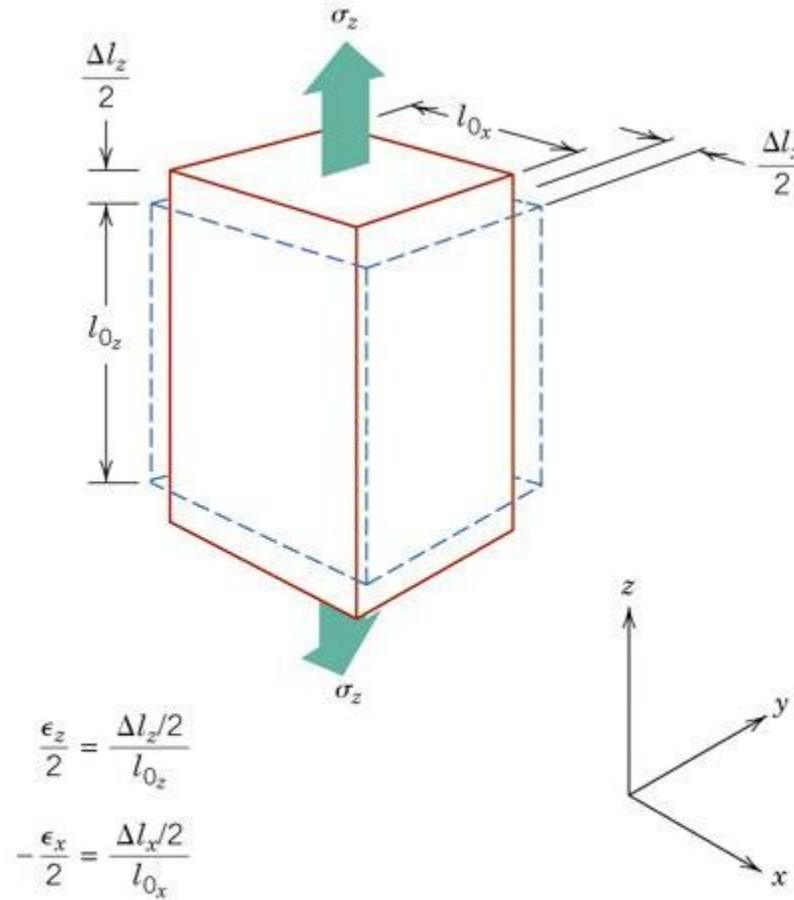
ceramics: $\nu \sim 0.25$

polymers: $\nu \sim 0.40$

Units:

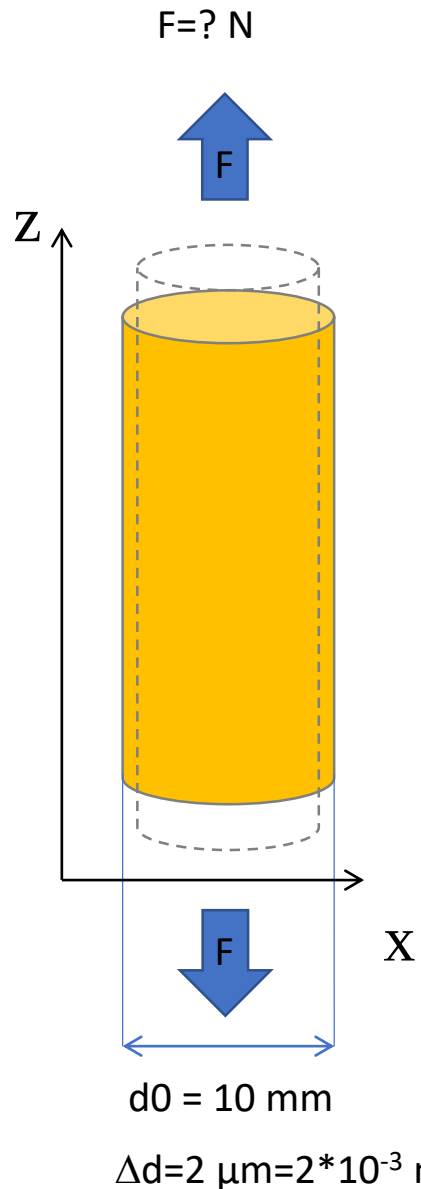
E : [GPa] or [psi]

ν : dimensionless



Exercise 2: You have a cylindrical aluminum component (diameter =10 mm, $\nu=0.34$, $E=70$ GPa): calculate the tensile load you need to apply to have a 2 micron smaller diameter component.

2- You have a cylindrical aluminum component (diameter =10 mm, $\nu=0.34$, $E=70$ GPa): calculate the tensile load you need to apply to have a 2 micron smaller diameter component.



$$\text{Poisson } (\nu) = - \epsilon_x / \epsilon_z$$

When a load F is applied the component elongates along z and reduce its diameter along x .

$$\epsilon_x = \frac{\Delta d}{d_0} = \frac{-2 \times 10^{-3} \text{ mm}}{10 \text{ mm}} = -2 \times 10^{-4}$$

(negative deformation for diameter reduction)

$$\epsilon_z = -\frac{\epsilon_x}{\nu} = -\frac{(-2 \times 10^{-4})}{0.34} = 5.8 \times 10^{-4}$$

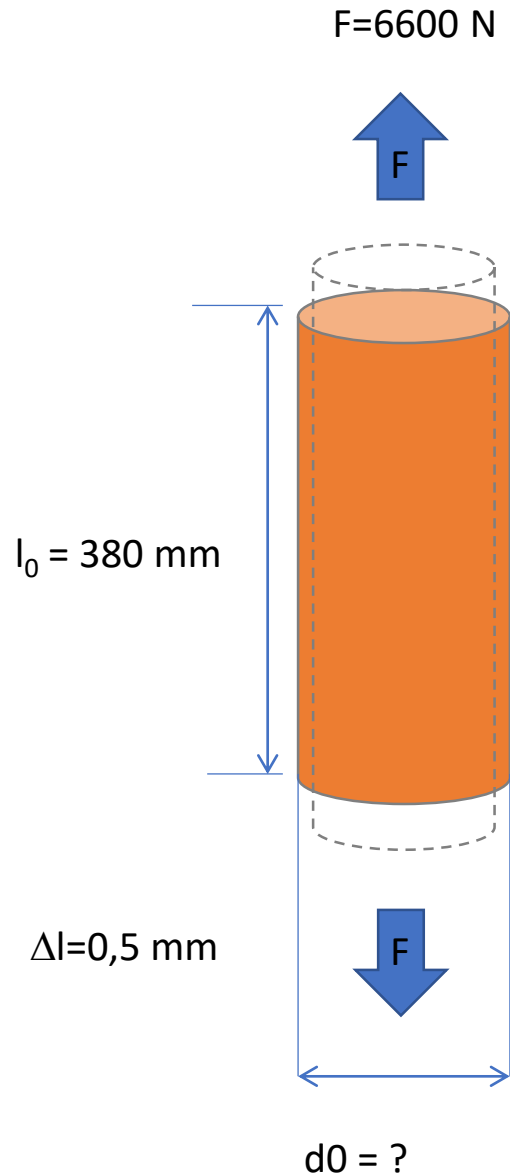
$$\sigma = \epsilon_z E = (5.8 \times 10^{-4}) (70 \times 10^3 \text{ MPa}) = 40.6 \text{ MPa}$$

$$A = \pi r^2 = 78.5 \text{ mm}^2$$

$$F = \sigma * A = 40.6 * 78.5 = \mathbf{3187 \text{ N}}$$

Exercise 3: Which is the diameter you need for a 380 mm long Cu component ($E=110$ GPa) to be loaded at 6600N, if you can accept an elongation of 0.5 mm.

3- Which is the diameter you need for a 380 mm long Cu component ($E=110 \text{ GPa}$) to be loaded at 6600N, if you can accept an elongation of 0.5 mm.



$$\varepsilon = \Delta l / l_0 = 0.5 / 380 = 0.0013$$

$$\sigma = E * \varepsilon = 110000 * 0.0013 = 144.7 \text{ Mpa}$$

$$\sigma = F / A$$

$$A = F / \sigma = 6600 / 144.7 = 45.6 \text{ mm}^2$$

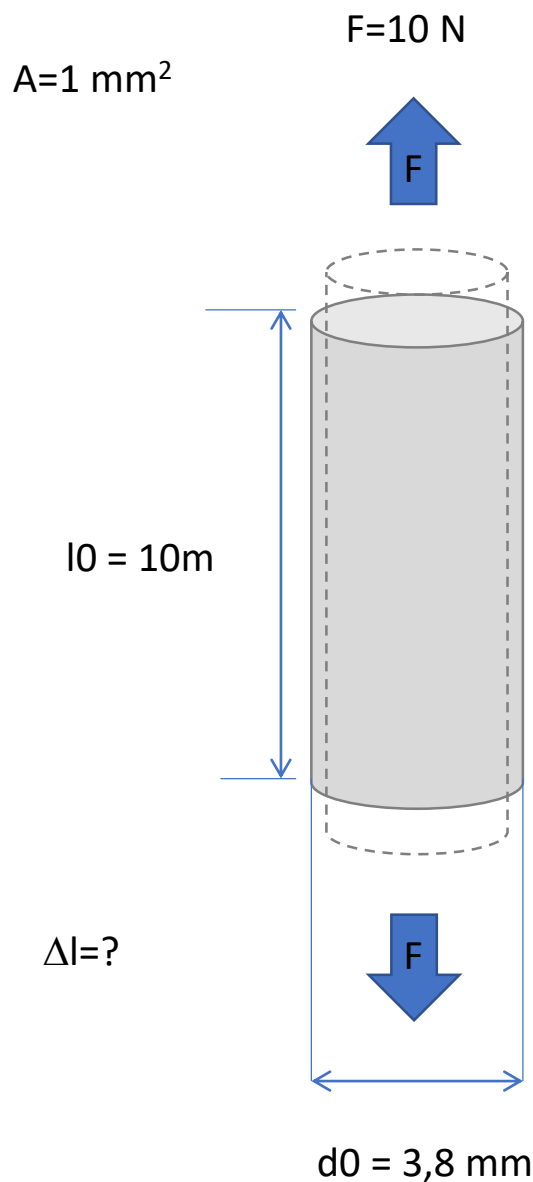
$$A = \pi * r^2$$

$$r = \sqrt{\frac{A}{\pi}} = 3.8 \text{ mm}$$

$$D = 2 * r = 7.6 \text{ mm}$$

Exercise 4: The elastic modulus of a steel is 193 GPa, of a bronze is 100 GPa, and of a polymer is 0,17 GPa ; calculate the elastic elongation of a 10 meter long wire made of steel, bronze or polymer , cross section is 1 mm², the applied load is F= 10 N.

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$$\varepsilon = \Delta l / l_0$$

$$\Delta l = \varepsilon * l_0$$

$$\varepsilon = \sigma / E$$

$$\sigma = F / A = 10 / 1 = 10 \text{ Mpa}$$

it is the same for all the materials

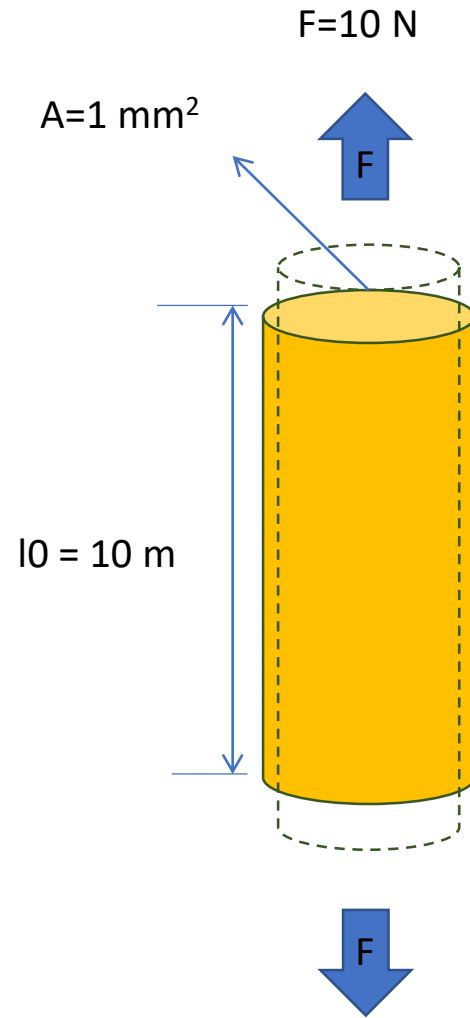
$$\varepsilon = \sigma / E \text{ depends on the Elastic Modulus (E), different for each material}$$

$$\varepsilon_{\text{steel}} = 10 / 193000 = 5.18 * 10^{-5} \rightarrow \Delta l_{\text{steel}} = 5.18 * 10^{-5} * 10000 = 0.518 \text{ mm}$$

$$\varepsilon_{\text{bronze}} = 10 / 100000 = 0.0001 \rightarrow \Delta l_{\text{bronze}} = 0.0001 * 10000 = 1 \text{ mm}$$

$$\varepsilon_{\text{polymer}} = 10 / 170 = 0.06 \rightarrow \Delta l_{\text{polymer}} = 0.06 * 10000 = 600 \text{ mm}$$

Exercise 5: A wire 10 m long with a cross section of 1mm^2 underwent tensile loading of 10N. Determine stress, strain and elongation if the wire is made of Copper ($E=110\text{ Gpa}$) or steel ($E=210\text{ Gpa}$). Compare different material on the available excel file.



$$\sigma = F/A = 10 \text{ [N]} / 1 \text{ [mm}^2\text{]} = \mathbf{10 \text{ MPa}}$$

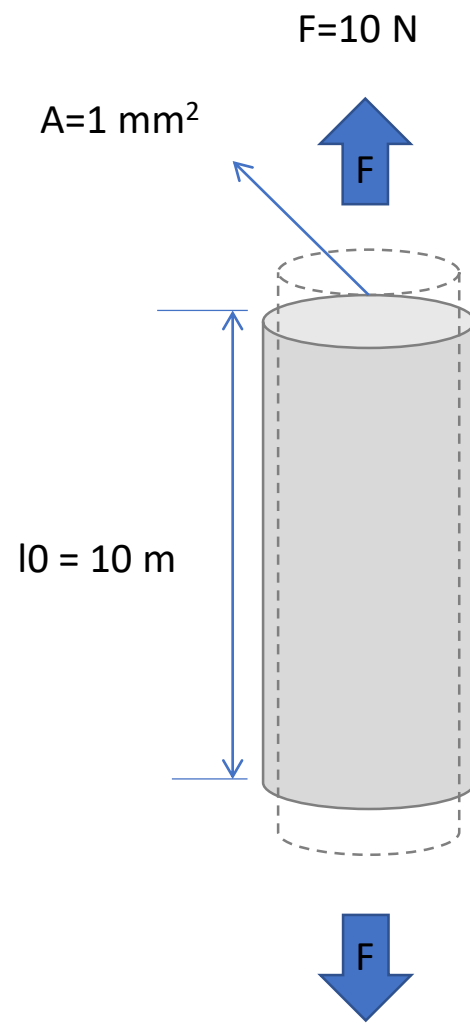
If the wire is made of **copper** ($E = 110 \text{ GPa}$) the elastic deformation is:

$$\sigma = E * \varepsilon \text{ cioe' } \varepsilon = \sigma / E$$

$$\varepsilon = 10 \text{ [MPa]} / 110\,000 \text{ [MPa]} = \mathbf{0.00009 = 0.009 \%}$$

$$\varepsilon = \Delta l / l_0 \quad \Delta l = \varepsilon * l_0$$

The total elongation on 10 m is
 $0.00009 \times 10 \text{ m} = 0.0009 \text{ m} = \mathbf{0.9 \text{ mm}}$



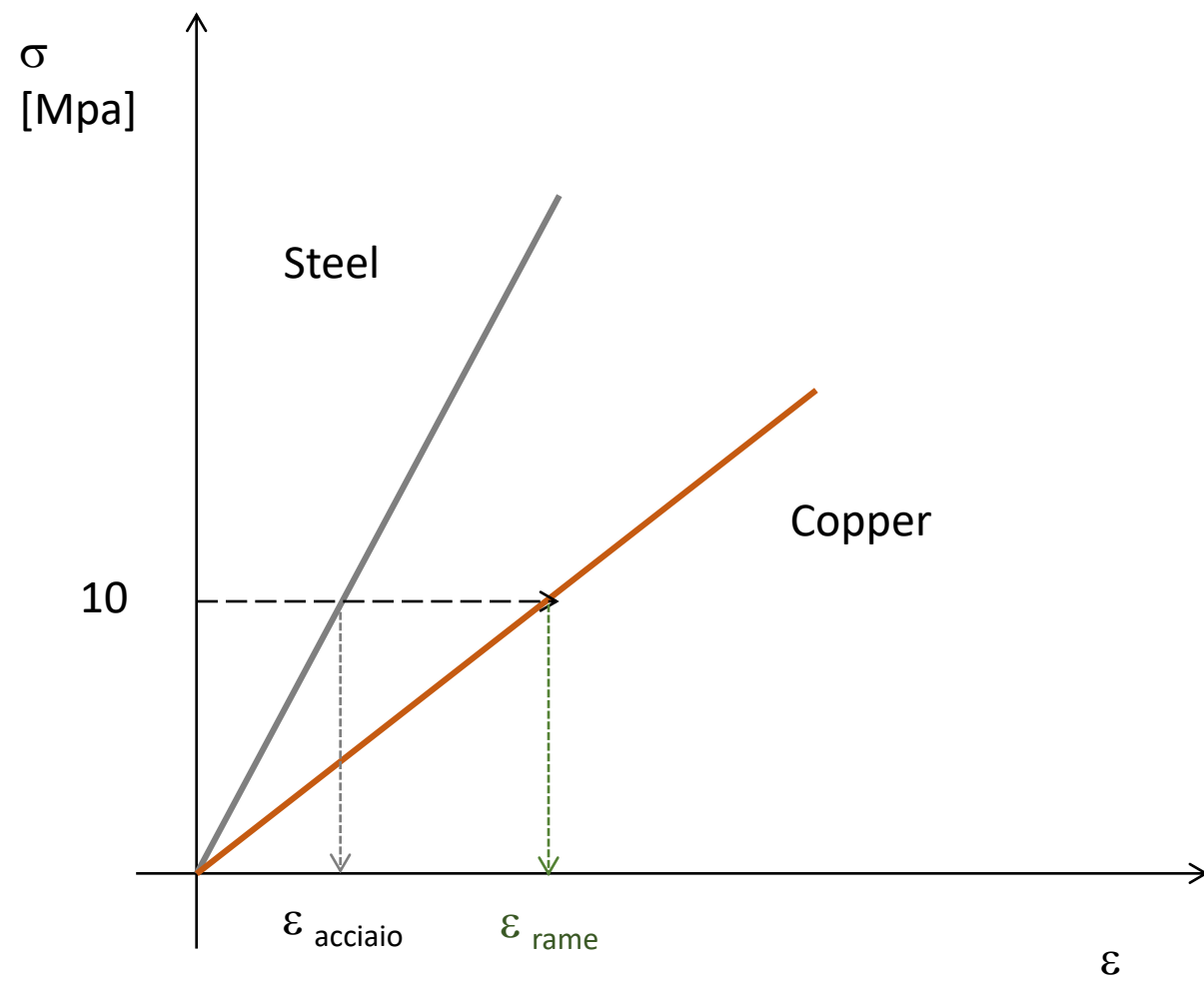
If the wire is made of **steel** ($E = 210 \text{ GPa}$):

$$\varepsilon = 10 / 210\,000 = 0.000048$$

The total elongation on 10 m is

$$0.000048 \times 10 \text{ m} = 0.00048 \text{ m} = \mathbf{0.48 \text{ mm}}$$

Since the elastic modulus of steel is almost double of the one of copper, the elastic deformation, in the same conditions, is almost a half.



Exercise 6

Consider an aluminium cylindrical sample with a diameter of 1cm.
the significant values for applied force underlined in traction test are:

- 4600 N (yield),
- 8450 N (maximum)
- 5700 N (rupture) .

When rupture occurs sample diameter is $\frac{1}{3}$ of the initial one

Determine:

- Yield strength
- Tensile strength
- Seemly fracture strength
- True fracture strength

Initial section: $(0,5)^2 \cdot \pi \text{ cm}^2 = 0,785 \cdot 10^{-4} \text{ m}^2$.

Yield strength:

$$\sigma_y = 4600 / 0,785 \cdot 10^{-4} = 5,86 \cdot 10^7 \text{ N/m}^2 = \mathbf{58.6 \text{ MPa}}$$

Tensile strength:

$$\sigma_m = 8450 / 0,785 \cdot 10^{-4} = 10,76 \cdot 10^7 \text{ N/m}^2 = \mathbf{107.6 \text{ MPa}}$$

Seemly fracture strength:

$$\sigma_R = 5700 / 0,785 \cdot 10^{-4} = 7,26 \cdot 10^7 \text{ N/m}^2 = \mathbf{72.6 \text{ MPa}}$$

If at the rupture sample present a diameter of 1/3 of the initial one, so 0.33cm, the real section that support the load is only:

$$(0,165)^2 \cdot \pi \text{ cm}^2 \Rightarrow 0,086 \cdot 10^{-4} \text{ m}^2$$

So the **true fracture strength** is:

$$\sigma_R = 5700 / 0,086 \cdot 10^{-4} = 66,3 \cdot 10^7 \text{ N/m}^2 = \mathbf{663 \text{ MPa}}$$

